

GAURI PRABHAKAR

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EDUCATION

Northeastern University

Master of Science in Artificial Intelligence

Boston, MA, USA

Expected Aug 2027

Chandigarh University

Bachelor of Engineering in Computer Science (AI and ML)

Chandigarh, India

Jun 2018 – May 2022

EXPERIENCE

NLP Teaching Assistant – Northeastern University, Boston, USA

May 2026 – Present

- Led lab sessions, office hours and graded assignments for **CS6120 NLP**, supporting **50+** graduate students under Prof. Amir Tahmasebi.

Supervisor – Northeastern University, EXP Makerspace, Boston, USA

May 2026 – Present

- Operate and supervise **8+** equipment categories, including 3D printers, laser cutters and woodworking tools, supporting **100+** patrons weekly.

Machine Learning Engineer – AYU, Pune, India

Oct 2024 – Jun 2025

- Architected end-to-end conversational AI for a wellness platform, owning **LLM** prompt design, custom **NLP** pipelines and sentiment-driven personalization from prototype to production; boosted session retention by **30%**.
- Engineered a low-latency model serving layer on async **FastAPI** with **LangChain** orchestration, integrating **OpenAI GPT** and **HuggingFace** with semantic response caching to handle **25k+** daily concurrent wellness conversations at production load.
- Drove a **40%** drop in response latency and **25%** lift in session engagement via context window optimization and reduced API calls.

Software Engineer – Schlumberger (SLB), Pune, India

Sep 2022 – Sep 2024

- Engineered high-performance **.NET/C#** APIs with query parsing over **MongoDB** aggregation pipelines and **Redis** master-data caching, achieving **80%** faster response times; mentored six engineers across **80+** APIs on a distributed, large-scale platform.
- Diagnosed cross-cutting href corruption in **APIGEE** API metadata and engineered a reusable shared flow policy, unblocking **50+** downstream consumers and recovering an estimated **\$100K** in infrastructure costs.
- Executed a zero-downtime migration of **327,000+** hierarchical MongoDB records across a full schema redesign, achieving zero data loss.
- Eliminated **RabbitMQ** message pile-ups across rig systems with automated correction pipelines, cutting daily queue incidents by **51%**.
- Built a **RAG** chatbot indexing internal documents, wiki pages and meeting transcripts into a **Pinecone** database for natural language querying.

Machine Learning Engineer – iSchoolConnect, Mumbai, India (Remote)

Feb 2022 – Aug 2022

- Productionized recommendation system access via an internal **Streamlit** tool for non-technical stakeholders, eliminating **85%** of cross-team bottlenecks and directly enabling **two** enterprise client onboardings worth **\$2M** in contracts.
- Diagnosed low recommendation accuracy and drove improvements through systematic feature engineering across student academic, financial and geographic profile dimensions, lifting recommendation accuracy by **18%**.
- Redesigned **MongoDB** schema and built a pipeline normalizing **20,000+** student records and ingesting **2,000+** scraped external data points.
- Automated MongoDB discrepancy detection that flagged data errors and fired instant developer alerts, eliminating manual daily **DB audits**.

PROJECTS

GPU Scheduler with Multi-Agent RL | Ray RLlib, MAPPO, CUDA, Kubernetes

Dec 2025 – Present

Trained **MAPPO** agents to schedule **CUDA**-accelerated GPU workloads across heterogeneous nodes, with a shared critic coordinating allocation; cut makespan by **26%** against a **Kubernetes** baseline.

Cryptographic Model Fingerprinting | Stable Diffusion XL, Llama 2/3, CleverHans, PyTorch

Aug 2025 – Mar 2026

Embedded cryptographic signatures into **Stable Diffusion XL** and **Llama 2/3** weights to trace AI content provenance; **78%** retention post-quantization, **71%** post-pruning, and signatures survived 3/5 **CleverHans** attack types.

Identifying RL Algorithms from Agent Behavior | Stable-Baselines3, scikit-learn

Jan 2026 – Apr 2026

Classifies whether an agent was trained with **DQN**, **PPO**, or **A2C** using trajectory data alone, with no access to model weights or training logs, reaching **87%** accuracy across **5** Gym environments on behavioral features including entropy, reward variance, and action autocorrelation.

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, Go, Bash

AI/ML: PyTorch, TensorFlow, HuggingFace, scikit-learn, LangChain, Ray RLlib, CUDA, NLP, Deep Learning, RL

Frameworks & Databases: FastAPI, .NET, MongoDB, PostgreSQL, Redis, Pinecone, MySQL, Streamlit, Flask, Linux

Tools & Infrastructure: Kubernetes, Docker, AWS, Azure, Git, CI/CD, RabbitMQ, Apigee, Postman, Agile/Scrum

OPEN SOURCE

Streamlit Login/Sign-Up UI | PyPI Library

Aug 2022 – Oct 2022

- Open-source auth library for Streamlit apps; **30,000+** downloads with special acknowledgement from the Streamlit team.

Algorithm Visualizer | Python, tkinter, pygame

Nov 2021 – Jan 2022

- Data structures visualization tool with dependency-free executable; research published in **IEEE**.